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Medium Eng./Hindi	ENGLISH	Registration Number	46104254
Center	ONLINE	Date	15 th /JULY/2026

	Maximum Marks	Marks Obtained
Q1		
Q2		
Q3		
Q4		
Q5		
Q6		
Q7		
Q8		
Q9		
Q10		
Q11		

Strongly Recommended

Plot No. 857, 1st Floor, Banda Bahadur Marg (Opp. Punjab & Sind Bank), Dr. Mukherjee Nagar, Delhi-110009

EVALUATION INDICATORS

1. Contextual Competence
2. Content Competence
3. Language Competence
4. Introduction Competence
5. Structure - Presentation Competence
6. Conclusion Competence

Overall Macro Comments / feedback / suggestions on Answer Booklet:

1.

2.

3.

4.

5.

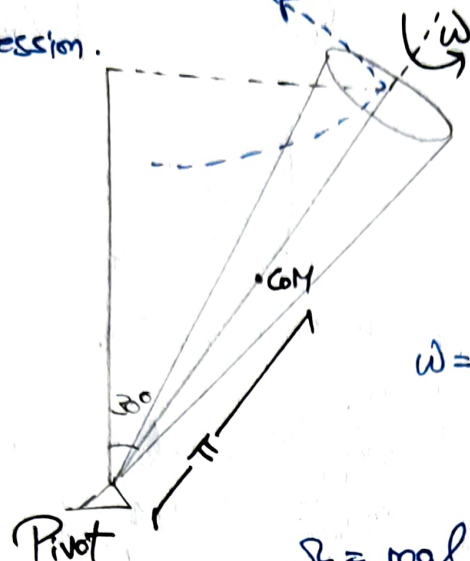
6.

All the Best

Q1. A top is spinning with 20 rev/s about an axis inclined at 30° with the vertical. Its radius of gyration is 5 cm. The centre of mass is x cm from the pivot point. Calculate the frequency of precession.

1)

Gyroscope
Precession.



$N = 20$ revolution per second.

$$\theta = 30^\circ$$

$$K = \sqrt{\frac{I}{M}} = 5 \text{ cm (Radius of gyration)}$$

$$\omega = 2\pi N = 2\pi(20) \text{ rad/s}$$

$$\omega = 125.66 \text{ rad/s} \quad \text{--- (1)}$$

$$\Omega = \frac{mgl}{I\omega} \quad \text{(Precessional angular velocity.)}$$

$$l = \pi \text{ cm} = 0.01 \times \pi \text{ m.} \quad \text{--- (2)}$$

$$\frac{I}{M} = 25 \times 10^{-4} \text{ m}^2$$

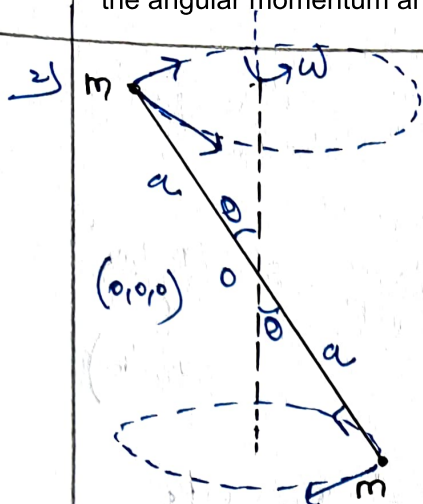
$$\therefore \Omega = \frac{gl}{\left(\frac{I}{M}\right)\omega} = \frac{(9.8)(\pi \times 10^{-2})}{25 \times 10^{-4} \times 125.66} = 0.98 \text{ rad/s.}$$

$$\text{Frequency of precession} = f = \frac{\omega \Omega}{2\pi} = 0.155 \text{ s}^{-1}$$

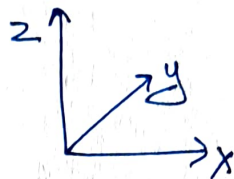
The orbital angular momentum of the Gyroscope about the vertical axis is neglected while establishing $\Omega = \frac{mgl}{I\omega}$ & the Torque of weight

causes the spin angular momentum to change.

Q2. Consider a dumbbell formed by two point masses m at the ends of a massless rod of length $2a$. It is constrained to rotate with angular velocity ω about an axis that makes an angle θ with the rod. Calculate the angular momentum and torque that is applied to the system.



choosing the below coordinate axis. $\vec{\omega} = \omega \hat{k}$



considering 'O' to be the origin of system.

The two point masses move in circles of radius

$$s = a \sin \theta$$

$$\text{From } \vec{L}_1 = \vec{r}_1 \times \vec{p}_1$$

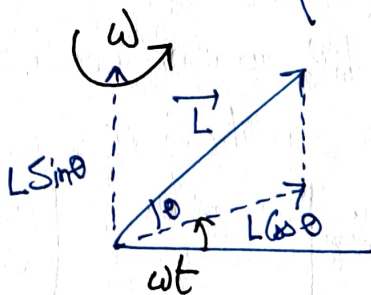
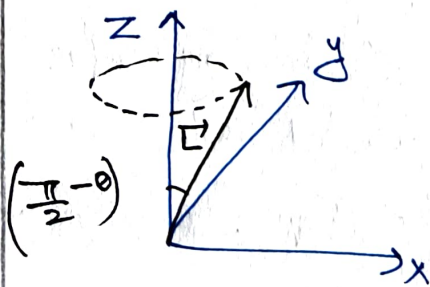
$$\& \vec{L}_2 = \vec{r}_2 \times \vec{p}_2$$

$$|\vec{L}_1| = a(m\omega s)$$

$$|\vec{L}_2| = a(m\omega s)$$

$$\therefore |\vec{L}| = |\vec{L}_1 + \vec{L}_2| = 2ma\omega s$$

The angular momentum of the system ($\vec{L}$) traces a 3D cone with semi-vertical angle as $\left(\frac{\pi}{2} - \theta\right)$



$$\vec{L} = L \sin \theta \hat{k} + L \cos \theta \cos \omega t \hat{i} + L \cos \theta \sin \omega t \hat{j}$$

where ωt is the angle rotated by $L \sin \theta$ in x - y plane in ' t ' time with angular velocity $\omega \hat{k}$.

$$\vec{L} = L \cos \theta (\cos \omega t \hat{i} + \sin \omega t \hat{j}) + L \sin \theta \hat{k}.$$

$$\& L = 2ma\omega\beta \quad \text{where } \beta = a \sin \theta$$

$$\therefore \vec{L} = 2ma^2\omega \sin \theta \cos \theta (\cos \omega t \hat{i} + \sin \omega t \hat{j}) + 2ma^2\omega \sin^2 \theta \hat{k}$$

Since the z -component of angular momentum is constant, $\overline{L}_z = 0$ i.e. $\left(\frac{d\vec{L}}{dt}\right)_z = 0$

$$\text{Now, } \frac{d\vec{L}}{dt} = \overline{\vec{L}} = 2ma^2\omega \sin \theta \cos \theta \left[-\omega \sin \omega t \hat{i} + \omega \cos \omega t \hat{j} \right]$$

$$\boxed{\overline{\vec{L}} = ma^2\omega^2 \sin 2\theta \left[-\sin \omega t \hat{i} + \cos \omega t \hat{j} \right]}$$

$$\boxed{|\overline{\vec{L}}| = m\omega^2 a^2 \sin 2\theta}$$

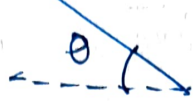
$$\boxed{|\vec{L}| = m\omega a^2 \sin 2\theta}$$

Q3. A uniform drum of radius b and mass M rolls without slipping down a plane inclined at an angle θ . Find the acceleration of the cylindrical drum along the plane.

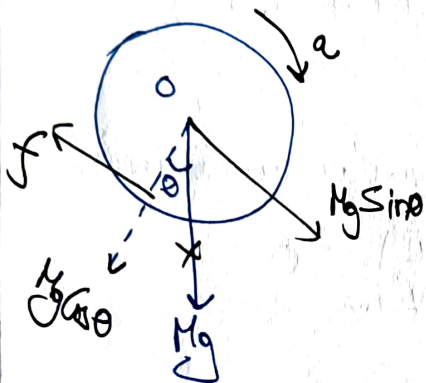
3)



pure rolling i.e. friction is present as rolling cannot occur without friction.



From Newton's 2nd law,
 $(\vec{F} = M\vec{a})$ along the incline



$$Mg \sin \theta - f = Ma \quad \text{--- (1)}$$

$$a = b\alpha \quad (\text{Condition for pure rolling}) \quad \text{--- (2)}$$

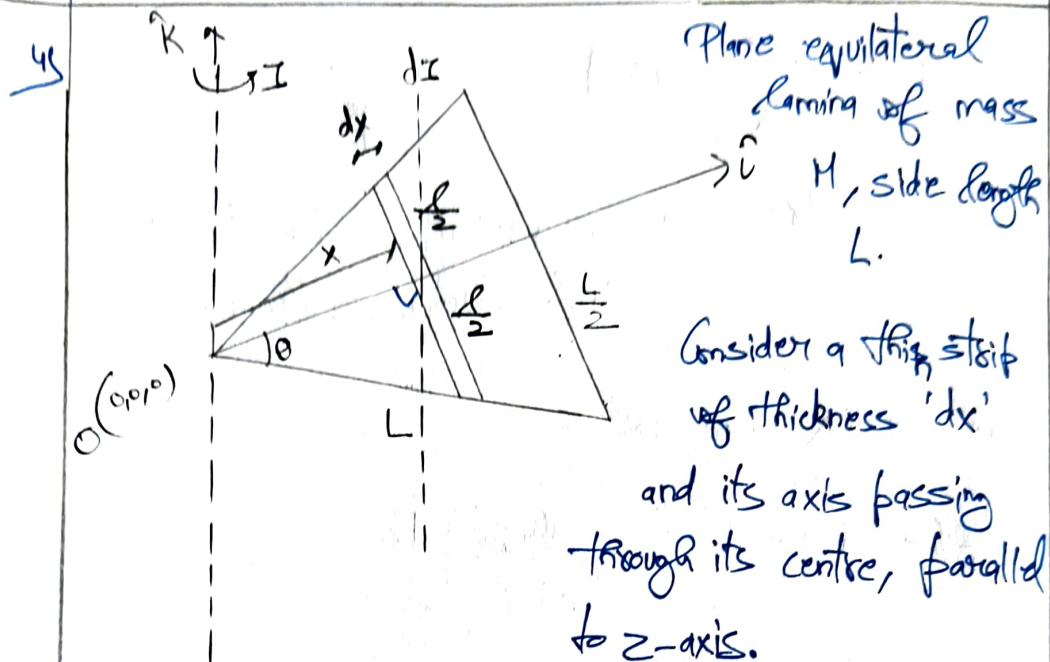
$$\sum \tau_o = I\alpha$$

$$\therefore f b = \left(\frac{Mb^2}{2} \right) \left(\frac{a}{b} \right) \therefore f = \frac{Ma}{2}$$

Substituting $f = \frac{Ma}{2}$ in (1), we get

$$Mg \sin \theta = \frac{3}{2} Ma \Rightarrow \boxed{a = \frac{2g \sin \theta}{3}}$$

Q4. Find the moment of inertia of a thin sheet of mass M in the shape of an equilateral triangle about an axis through a vertex and perpendicular to the sheet. The length of each side is L .



Moment of inertia of the thin strip = dI
about an axis parallel to R

$$\therefore \boxed{I = \int dI'}$$

Now, using parallel axis theorem for the infinitesimal strip, $\boxed{dI' = dI + (dm)x^2}$ — (1)

where $dI' =$ moment of inertia of the strip about z -axis.

$$\therefore dI' = \frac{(dm)L^2}{12} + (dm)x^2 \quad ; \quad L = \text{length of strip}$$

$$\tan \theta = \frac{1}{\sqrt{3}} = \frac{L/2}{x} \quad \therefore \frac{L}{2x} = \frac{1}{\sqrt{3}} \quad \therefore \boxed{L = \frac{2x}{\sqrt{3}}}$$

Substituting $L = \frac{2x}{\sqrt{3}}$ in eqn (1), we get,

$$dI' = dm \left(\frac{l^2}{12} + x^2 \right), \quad l = \frac{2x}{\sqrt{3}}$$

$$dm = (\sigma) l dx, \quad \sigma = \text{Surface mass density of triangular lamina,}$$

$$dm = \sigma l dx \text{ in } \odot \neq$$

$$\int dI' = I = \int \sigma l \left(\frac{l^2}{12} + x^2 \right) dx$$

$$I = \sigma \int \left(\frac{l^3}{12} + l x^2 \right) dx$$

$$= \sigma \left[\int_0^{L/\sqrt{3}} \left(\frac{2x}{\sqrt{3}} \right)^3 \cdot \frac{1}{12} + \left(\frac{2x}{\sqrt{3}} \right) x^2 dx \right]$$

$$I = \sigma \int_0^{\frac{\sqrt{3}L}{2}} \frac{8x^3}{3\sqrt{3} \cdot 12} + \frac{2x^3}{\sqrt{3}} dx$$

$$I = \sigma \left(\frac{2}{9\sqrt{3}} + \frac{2}{\sqrt{3}} \right) \cdot \int_0^{\frac{\sqrt{3}L}{2}} x^3 dx$$

$$= \frac{10}{9} \left(\frac{2}{\sqrt{3}} \right) \sigma \int x^3 dx$$

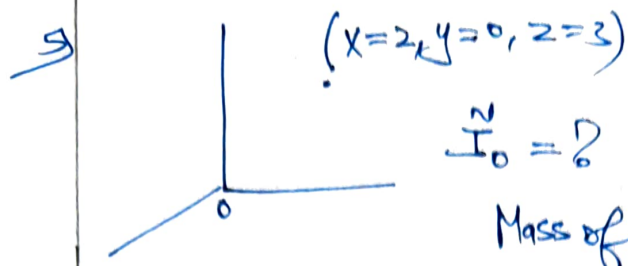
$$I = \frac{20}{9\sqrt{3}} \sigma \cdot \frac{1}{4} \cdot \left(\frac{\sqrt{3}L}{2} \right)^4 = \frac{5\sigma}{4\sqrt{3}} \cdot \frac{L^4}{16} = \frac{5}{16\sqrt{3}} \sigma L^4$$

Now, $\sigma = \frac{M}{\frac{\sqrt{3}L^2}{4}}$ Now, σ

$$I = \frac{5}{16\sqrt{3}} \cdot L^4 \cdot \frac{M}{\frac{\sqrt{3}L^2}{4}} = \frac{5}{4 \times 3} ML^2 \Rightarrow$$

$$I = \frac{5}{12} ML^2$$

Q5. A particle of mass m is located at $x = 2$, $y = 0$, $z = 3$. Find the moments and products of inertia relative to the origin.



$$I_{xx} = m(y^2 + z^2) = m(0 + 9) = 9m$$

$$I_{yy} = m(x^2 + z^2) = m(4 + 9) = 13m$$

$$I_{zz} = m(x^2 + y^2) = 4m$$

$$I_{xy} = -mxy = -m(2)(0) = 0 = I_{yx}$$

$$I_{yz} = I_{zy} = -myz = -m(0)(3) = 0$$

$$I_{zx} = I_{xz} = -mzx = -6m$$

$$\therefore \vec{I} = \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} = \begin{pmatrix} 9 & 0 & -6 \\ 0 & 13 & 0 \\ -6 & 0 & 4 \end{pmatrix} m$$

The negative sign in the product of inertia is indicative of the geometry of vector triple product and the coupling that tilts the angular momentum vector away from the spin direction.

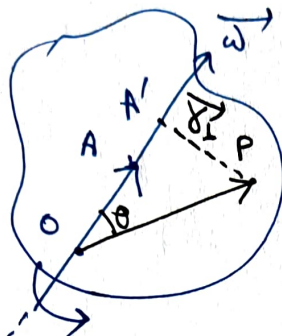
I_{xy} & $I_{yz} = 0 \Rightarrow$ The body is symmetric
about the xy plane & yz plane.

$$\boxed{\vec{L} = \vec{I}\omega}$$

Q6. A rigid body is spinning with angular velocity 4 rad/s about an axis parallel to the direction $(4i - 3k)$, passing through the point A with $OA = 2i + 3j - k$, where O is the origin. Find the magnitude and direction of the linear velocity of the body at point P with $OP = 4i - 2j + k$.

$$\vec{\omega} = \omega \hat{\omega} = 4(\hat{\omega})$$

$$\hat{\omega} = \frac{\vec{\omega}}{|\vec{\omega}|} = \frac{4\hat{j} - 3\hat{k}}{5} \quad \therefore \boxed{\vec{\omega} = \frac{16}{5}\hat{j} - \frac{12}{5}\hat{k}}$$



$$\vec{OA} = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$\vec{OP} = 4\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{v}_P = \vec{\omega} \times \vec{r}$$

$$|\vec{v}_P| = \omega (\delta_L \sin \theta)$$

$$\delta_L = \delta \sin \theta$$

where δ_L = perpendicular distance of P from the axis of rotation.

$$\cos \theta = \frac{\vec{OA} \cdot \vec{OP}}{|\vec{OA}| |\vec{OP}|} = \frac{(2\hat{i} + 3\hat{j} - \hat{k}) \cdot (4\hat{i} - 2\hat{j} + \hat{k})}{\sqrt{14} \sqrt{21}}$$

$$= \frac{8 - 6 - 1}{\sqrt{294}} = \frac{1}{\sqrt{294}} \quad \theta = \cos^{-1} \left(\frac{1}{\sqrt{294}} \right)$$

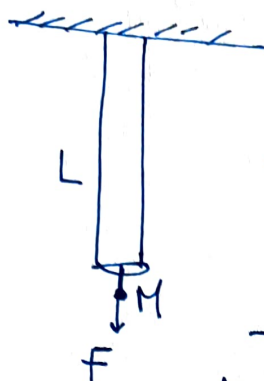
$$\boxed{\theta = 86.65^\circ} \quad \therefore |\vec{OP}| = \sqrt{21}$$

$$|\vec{v}_P| = (4)(\sqrt{21}) \sin(86.65^\circ)$$

$$\boxed{|\vec{v}_P| = 18.29 \text{ m/s}}$$

Q7. A mass M is suspended from an elastic string so that pulling the mass down and releasing it causes SHM. If an additional mass m is added, the period of oscillation changes in the ratio 5 : 4. Find m/M .

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Within the proportional limit of stress-strain curve,

stress $\propto$ strain & longitudinal stress,

$$Y = \frac{\text{Stress}}{\text{Strain}} \quad \text{where } Y = \text{Young's modulus.}$$

$$Y = \frac{\frac{F}{A}}{\frac{\Delta x}{L}} \quad \therefore \frac{FL}{A\Delta x} = Y \propto F = \left(\frac{AY}{L}\right)\Delta x$$

$F \propto \Delta x$ (Restoring force resulting into small oscillations)

$$T = 2\pi \sqrt{\frac{ML}{AY}} \quad \therefore T_1 = 2\pi \sqrt{\frac{ML}{AY}}$$

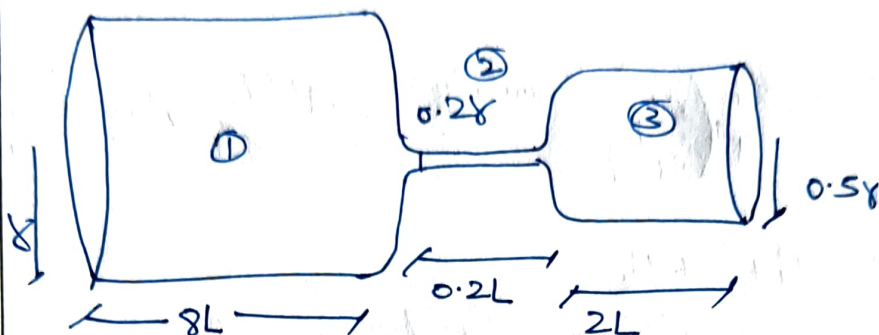
Now, M is increased to $(M+m)$, updating the period of oscillation to $T_2 = 2\pi \sqrt{\frac{(M+m)L}{AY}}$

$$\therefore \frac{T_2}{T_1} = \sqrt{\frac{(M+m)}{M}} \Rightarrow \left(\frac{5}{4}\right) = \sqrt{\frac{M+m}{M}}$$

$$\propto \frac{25}{16} = 1 + \left(\frac{m}{M}\right) \quad \therefore \frac{m}{M} = \frac{9}{16}$$

Q8. Three capillaries of lengths $8L$, $0.2L$ and $2L$ with radii r , $0.2r$ and $0.5r$ are connected in series. If the total pressure across the system is P , deduce the pressure across the shortest capillary.

8) The capillaries connected in series.



Total pressure across the system $= P$.

To find $P_2 =$ Pressure difference across the shortest capillary.

From Poiseuille's equation of volumetric flow rate,

$$Q = \frac{P \pi R^4}{8 \eta L'}$$

This is the fluid analog of the electrical ohm's law. $V = IR$.

where, Pressure diff. $\equiv$ Voltage.

Q (flow rate) $\equiv$ Current I

$\frac{8 \eta L'}{\pi R^4} \equiv$ fluid resistance or R .

In series connection, the flow through the

capillaries remains same i.e. $Q_1 = Q_2 = Q_3$ (2)

& $P = P_1 + P_2 + P_3$ (1)

$$Q = \frac{P_1 \pi r_1^4}{8\eta l_1}, \quad Q = \frac{P_2 \pi r_2^4}{8\eta l_2}, \quad Q = \frac{P_3 \pi r_3^4}{8\eta l_3}$$

$$\therefore P_1 = \frac{Q 8\eta}{\pi} \left(\frac{l_1}{r_1^4} \right) \text{ Similarly } P_2 = K \left(\frac{l_2}{r_2^4} \right), P_3 = K \left(\frac{l_3}{r_3^4} \right)$$

$$\therefore P = P_1 + P_2 + P_3$$

$$P = K \left[\frac{l_1}{r_1^4} + \frac{l_2}{r_2^4} + \frac{l_3}{r_3^4} \right]$$

$$P = \frac{8\eta Q}{\pi} \left[\frac{8L}{r^4} + \frac{0.2L}{(0.2)^4 r^4} + \frac{2L}{(0.5)^4 r^4} \right]$$

$$P = \frac{8\eta Q}{\pi} \left[8 + \frac{1}{(0.2)^3} + \frac{2}{(0.5)^4} \right] \frac{L}{r^4} = \boxed{\frac{1320 \eta Q L}{\pi r^4}}$$

— (2)

$$\text{Now, } P_2 = \frac{8\eta l_2 Q}{\pi r_2^4} = \frac{8\eta (0.2)L Q}{\pi (0.2)^4 r^4}$$

$$P_2 = \frac{8\eta L Q}{\pi r^4} (125) = (1000) \frac{\eta L Q}{\pi r^4}$$

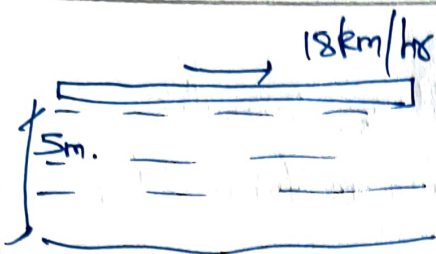
$$\text{Now, } \frac{\eta L Q}{\pi r^4} = \frac{P}{1320}, \text{ substituting in the above}$$

$$\text{eqn } \Rightarrow P_2 = (1000) \left(\frac{P}{1320} \right) \approx 0.75P$$

$$\boxed{P_2 = \frac{25P}{33}} \propto \boxed{0.758P}$$

Q10. The velocity of water in a river is 18 km/h near the surface. If it is 5 m deep, find the shearing stress between horizontal layers of water. The coefficient of viscosity of water = 10^{-2}

10)



Considering the flow to be laminar i.e. water layers glide over each other without intermixing with a velocity less than the critical velocity.

Given $\eta = 10^{-2}$ poise,

Depth of river, $z = 5$ m.

$$\therefore \text{linear velocity gradient} = \left(\frac{\Delta v}{\Delta z} \right) = - \frac{18 \times \frac{5}{18} \text{ m/s}}{5}$$

$$= -1 \text{ s}^{-1}$$

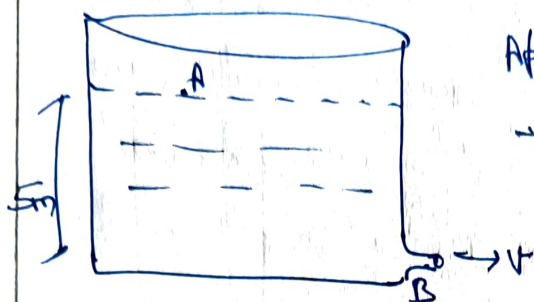
$$\text{Now, viscous force } F_v = -\eta A \frac{dv}{dz} \Rightarrow$$

$$F_v = -(10^{-2}) A (-1)$$

$$\therefore \boxed{\sigma = \text{Shearing Stress between the layers} = \frac{F_v}{A} = 10^{-2} \frac{\text{N}}{\text{m}^2}}$$

$$\boxed{\sigma = 0.01 \text{ N/m}^2}$$

Q11. A large tank open to the atmosphere is filled with water to a height 5 m. A tap near the bottom is opened. Determine the water velocity at the outlet.



Applying Bernoulli's
theorem b/w A & B,

$$P_0 + \frac{1}{2} \rho v_1^2 + \rho gh = P_0 + \frac{1}{2} \rho v^2 + 0$$

Since the atmospheric pressure is same at A & B.

$$\boxed{\frac{v_1^2}{2} + gh = \frac{v^2}{2}} \quad \text{--- (1)}$$

Now, applying equation of continuity $A_1 v_1 = A_2 v_2$

$$\therefore \boxed{v_1 = \frac{A_2 v}{A_1}}$$

Substituting in (1) $\Rightarrow$

$$\frac{A_2^2 v^2}{A_1^2} + gh = \frac{v^2}{2}$$

$$\therefore v^2 \left[\frac{1}{2} - \left(\frac{A_2}{A_1} \right)^2 \right] = gh$$

$$\boxed{v = \frac{\sqrt{2ghA_1^2}}{\sqrt{A_1^2 - A_2^2}} = \sqrt{\frac{2gh}{1 - \left(\frac{A_2}{A_1} \right)^2}}}$$

if the area of orifice $A_2 \ll A_1$

$$\text{i.e. } \frac{A_2}{A_1} \approx 0$$

$$\therefore \boxed{v = \sqrt{2gh}} = \sqrt{2 \times 9.8 \times 5} \text{ m/s}$$

$$\boxed{v = 7\sqrt{2} \text{ m/s} = 9.89 \text{ m/s}}$$